

CS 593

Natural Language Processing

Fall 2025
Instructor: Jonathan Rusert



Introductions - Me

Academics

University of Minnesota, Duluth

- B.S. Computer Science, Mathematics
- M.S. Computer Science

University of Iowa

- PhD Computer Science

Purdue Fort Wayne

- Visiting Professor 2022
- Assistant Professor 2023

Introductions - Me

Research

- Natural Language Processing, Obfuscation, Text Classification, NAACL Adversarial Attacks, 2024 Mexico City Privacy



V Attack: Taking advantage of Text Classifiers' horizontal vision
Jonathan Rusert
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Abstract

Text classification systems have continuously improved in performance over the years. However, nearly all current SOTA classifiers have a similar shortcoming, they process text in a horizontal manner. Vertically written words will not be recognized by a classifier. In contrast, humans are easily able to recognize and read words written both horizontally and vertically. Hence, a human adversary could write problematic words vertically and the meaning would still be preserved to other humans. We simulate such an attack, VerAttack. VerAttack identifies which words a classifier is reliant on and then rewrites those words vertically. We find that VerAttack is able to grossly drop the accuracy of 4 different transformer models on 5 datasets. For example, on the SST2 dataset, VerAttack reduces the accuracy from 88.8% to 78.8%.

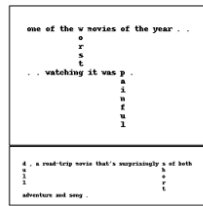


Figure 1: Examples of texts perturbed by VerAttack.

Rethinking Attack: Testing the Reliability of Attack Detection

Jonathan Rusert
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Abstract

In response to adversarial text attacks, attack detection models have been proposed and shown to successfully identify text modified by adversarial attacks. Attack detection models can be leveraged to provide an additional check for NLP models and give signals for human signs. However, the reliability of these models has not been thoroughly explored. Thus, we propose and test an attack, Rethinking. Rethinking aims to make attack detection models unreliable by modifying a text to cause the detection model to predict an attack, while keeping the classifier correct. This causes a misuse to misuse the classifier and detection. If a human sees that the detector is giving an "incorrect" prediction, the classifier is correct, then the human will use the detector as unreliable. We test this worst-case model on 4 datasets against 7 detection-detecting classifiers. We find that Rethinking is able to cause the detection model to predict an attack, while keeping the classifier correct. It is accurate between 30-71 points, while maintaining the accuracy of the classifier. As an initial defense, we propose a simple text check which requires no retraining of the classifier or detector and increases detection accuracy greatly. This novel check often outperforms state-of-the-art adversarial model detection models.

and privacy (Wu and Wang, 2022). In response, defenses have been researched (Li et al., 2021b, 2022; Wang et al., 2022). Defenses often rely on the assumption that the attacker will not be able to modify the text to cause the classifier to predict an attack. While defenses improved the robustness of classifiers, they often require retraining of models (Wu et al., 2020; Wang et al., 2022) or increase the number of queries required to make a classification (Zeng et al., 2021; Rusert and Vasanthan, 2022). Thus, attack detection helps as an additional line of defense for classifiers.

Attack detection algorithms aim to classify whether a text has been modified by an adversarial algorithm to subvert a classifier (Bhunia et al., 2021; Moses et al., 2022). These algorithms have been proposed to help guide and potentially modify text to human intermediaries. Thus, a reliable model leverage these algorithms to flag text for humans to check against the automated classifier. It potential problems arise if the detection algorithm flags too many texts incorrectly, as this will cause human intermediaries' work load to greatly increase.

In this research, we aim to investigate the reliability of text attack detection algorithms to attacks which cause distrust between humans and the algorithms. Thus, we propose an attack, Rethinking, which causes the attack detection system to trigger without causing the classifier to fail. This type of attack is different from normal adversarial attacks in terms which only focus on failure of the NLP system. It is similar to other types of security attacks

1. Introduction

Recent work on text and related problems continues to grow, competition has increasingly become

BinarySelect to Improve Accessibility of Black-Box Attack Research

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Abstract

Adversarial text attack research is useful for testing the robustness of NLP models, however, the rise of transformers has greatly increased the time required to test attacks. Especially when researchers do not have access to adequate resources (e.g. GPUs). This can hinder attack research, as modifying one example for an attack can require hundreds of queries to a model, especially for black box attacks. Often these attacks remove one token at a time to find the ideal one to change, requiring n queries (the length of the text) right away. We propose a more efficient selection method called BinarySelect.

(Sadriadeh et al., 2022; Choi et al., 2021) have access to a model's weights and architecture. This allows the attack to more quickly find the tokens which the classifier is leveraging, however, this may be unrealistic when considering models deployed online. Black box attacks (Deng et al., 2022; Le et al., 2022) have access only to the output (e.g. label) and probabilities (or logits) of a model. This restriction means that black box attacks spend more time querying the model to find the same tokens. In the case of text classification, attacks often remove or mask one token (word) at a time and check the change in probability (Jin et al., 2020; Li et al.,

Overcoming Black-box Attack Inefficiency with Hybrid and Dynamic Select Algorithms

Abhinav Shankar Belde and Rohit Ramkumar and Jonathan Rusert
Department of Computer Science,
Purdue University, Fort Wayne

Abstract

Adversarial text attack research plays a crucial role in evaluating the robustness of NLP models. However, the increasing complexity of transformer-based architectures has dramatically raised the computational cost of attack testing, especially for researchers with limited resources (e.g. GPUs). Existing popular black-box attack methods often require a large number of queries, which can make them inefficient and impractical for researchers. To address these challenges, we propose two new attack selection strategies called Hybrid and Dynamic Select, which better combine the strengths of previous selection algorithms. Hybrid Select merges generalized BinarySelect techniques with GreedySelect by introducing a size threshold to decide which selection algorithm to use. Dynamic Select provides an alternative approach of combining the generalized Binary and GreedySelect by learning which lengths of texts each selection method should be applied to. This greatly reduces the number of queries needed while maintaining attack effectiveness (a limitation of BinarySelect). We also extend this to a sentence level, and find that our method is able to reduce the number of required queries per attack up to 16.7% on

highest probability drop. GreedySelect achieves high precision. However, its word-by-word approach often results in high query counts, limiting its efficiency.

As NLP models continue to grow in size, high query counts can translate to attacks taking a large amount of time per text. This inefficiency further affects researchers or users who do not have access to high end GPU's to carry out attacks. Furthermore, an increased time in attack execution, reduces the ability for researchers to adequately test their models.

To address these challenges, BinarySelect (Ghosh and Rusert, 2025) adopted a divide-and-conquer strategy inspired by binary search. By partitioning the input into two segments and evaluating the probability change caused by excluding each segment, BinarySelect aimed to narrow down the search space when selecting an influential word. While BinarySelect helped reduced the number of queries compared to GreedySelect in some instances, it caused a drop in attack effectiveness. Building on these foundations, this paper proposes hybrid approaches that integrate query-efficient methods like BinarySelect with GreedySelect's effectiveness. These strategies aim to bal-

Introductions - Cat

Belle



Class Information

Lecture:

- Instructor: Jonathan Rusert
- Office: ET 125K
- Office Hours: M 3 – 4, T 12 – 1, W 1 – 3 PM or by appointment
- Email: jrusert@pfw.edu
- Course Website: purdue.brightspace.com

Course Goals

- Become familiar Natural Language Processing (NLP) and its problems
 - Q&A
 - Classification
 - NLG
 - Transformers
 - ...
- Think through ways of addressing of basic and more difficult NLP problems

Why Study NLP?

Why Study NLP?

- Conversational Agents (Alexa, Siri, Google)
- Automatic Classification
- Obfuscation
- Defending against Adversarial Attacks
- Fairness and Bias
- Natural Language Generation
 - GPT
 - LLMs
- Many areas of research (<https://www.2022.aclweb.org/papers>)

Student Goals

External:

- Develop ways to approach NLP problems
- Strengthen ability to examine and analyze text

Internal:

- Improve algorithmic thinking
- Obtain self-sense of accomplishment

Textbook

Speech and Language Processing

An Introduction to Natural Language Processing,
Computational Linguistics, and Speech Recognition

Third Edition draft

Daniel Jurafsky
Stanford University

James H. Martin
University of Colorado at Boulder

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<https://web.stanford.edu/~jurafsky/slp3/>

Grading

Item	Percentage	Points
Homework	30	(6 at 50) = 300
Quizzes	24	(3 at 80) = 240
Final Project	40	400
Attendance/Class Participation	6	60
Total	100	1000

Grading

- A 90 - 100
- B 80 - 89
- C 70 - 79
- D 60 - 69
- F Below 60

Class Policies

- No late assignments accepted*
 - * homeworks considered late when the grader starts grading the assignment
 - (guaranteed grader won't start grading before the due date)
- Email or contact about special circumstances (e.g. extended sickness, accident, etc.)

Class Policies

- Homework done **individually**
 - Come to office hours if stuck
- Academic Integrity Code in Purdue University Fort Wayne student manual
 - Cheating
 - Plagiarism
- Class participation
- Late Arrival/Attendance

Class Policies

- Read assigned chapters and paper(s) before class
 - Bring questions ready to discuss
- Class time
 - Focus on complicated aspects
 - Work through examples
 - Answer questions
 - Don't be afraid to ask questions

Class Policies

- Be considerate of others
- Help to create a positive learning environment for yourself and others
 - Cell phones off/silenced, no extra conversations, etc.
 - Participate in discussions
 - Ask questions
 - What if ...?
 - Why is it that ...?
 - Could you explain this notion one more time?

What to Expect

Combination of introduction NLP concepts

- Textbook and homeworks, examples in class

And examination of State-of-the-art solutions

- Recently published research papers

How to achieve a satisfactory grade

- Access textbook
- Read Research Papers and participate in discussions
- Attend class
- Understand class slides and examples
 - Rework through after class if difficulty during
- Understand HW problems and solutions
- Form study group
 - Discuss concepts from readings
 - Explain concepts to others
- Ask me
- Don't wait until the last minute!
- Review materials frequently

I am here to help!

- You are here to learn, not already have all the answers
 - No shame in asking questions
 - In class
 - Office Hours
- Each semester students struggle and refuse to ask for help
 - The earlier you ask, the better you will be prepared for the rest of the course
- I want to help, but cannot force you to learn
- My office hours are specifically set to help my students
 - If there are none that work for you, email me to set up other times

Q & A